On a sequential basis, ADG revenues were up 3.3%, driven by an increase in volumes of approximately 8%, partially offset by a decrease in average selling prices of approximately 5%, mostly attributable to product mix. AMS revenues increased 12.1% driven by Analog and Imaging products. AMS increase was due to an increase of approximately 5% in average selling prices, entirely due to product mix, and to higher volumes of approximately of 7%. MDG revenues increased by 7.9%, mainly driven by Microcontrollers, due to both higher average selling prices of approximately 6%, entirely due to product mix, and higher volumes of approximately 2%.

On a year-over-year basis, fourth quarter net revenues increased by 4.0%. ADG revenues decreased 4.5% compared to the year-ago quarter on lower revenues in both Automotive and Power Discrete. The decrease was entirely due to lower average selling prices of approximately 4%, while volumes remained substantially flat. The decrease in average selling prices was a combination of less favorable product mix and lower selling prices. AMS fourth quarter revenues grew 9.9% year-over-year, mainly driven by Analog and Imaging. The increase was entirely due to higher average selling prices of approximately 18%, entirely attributable to product mix, partially offset by lower volumes of approximately 8%. MDG fourth quarter revenues increased by 7.6%, mainly driven by Microcontrollers. The increase was due to higher average selling prices of approximately 9%, entirely due to improved product mix.

Net Revenues by Market Channel[1]

	Three Months Ended		
	December 31, 2019	**September 29, 2019**	**December 31, 2018**
	(Unaudited, in %)		
OEM	72%	72%	69%
Distribution	28	28	31
Total	**100%**	**100%**	**100%**

(1) Original Equipment Manufacturers ("OEM") are the end-customers to which we provide direct marketing application engineering support, while Distribution customers refers to the distributors and representatives that we engage to distribute our products around the world.

By market channel, our fourth quarter revenues in Distribution amounted to 28% of our total revenues, flat compared to the previous quarter and decreasing on a year-over-year basis.

Net Revenues by Location of Shipment[1]

	Three Months Ended			% Variation	
	December 31, 2019	**September 29, 2019**	**December 31, 2018**	**Sequential**	**Year-Over-Year**
	(Unaudited, in millions)				
EMEA	$ 538	$ 549	$ 617	(2.0)%	(12.8)%
Americas	360	326	342	10.5	5.4
Asia Pacific	1,856	1,678	1,689	10.6	9.9
Total	**$ 2,754**	**$ 2,553**	**$ 2,648**	**7.9%**	**4.0%**

(1) Net revenues by location of shipment are classified by location of customer invoiced or reclassified by shipment destination in line with customer demand. For example, products ordered by U.S.-based companies to be invoiced to Asia Pacific affiliates are classified as Asia Pacific revenues.

By region of shipment, in the 2019 fourth quarter, Asia Pacific revenues grew sequentially by 10.6%, principally in Imaging, Analog and Microcontrollers, Americas was up by 10.5%, mainly driven by Digital and Power Discrete, and EMEA decreased by 2.0%.

On a year-over-year basis, revenues grew 9.9% in Asia Pacific, mainly due to Analog, Microcontrollers and Imaging. In Americas, revenues grew 5.4%, mainly driven by Digital and Power Discrete, while it decreased by 12.8% in EMEA, mainly due to lower sales in Automotive, Power Discrete and Analog.